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Assignment 3, Part 1

Suppose we observe $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, where we assume the following linear relation

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \in \{1, 2, \dots, n\}$$

with σ^2 being constant across all values of x_i (i.e. *homoskedasticity*).

We fit a line using the method of *ordinary least squares* (OLS) and want to estimate the value on the line at a new point x_0 . Denoting the value on the line by μ_0 , the estimate is

$$\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

- Please derive the expression for the variance of $\hat{\mu}_0$ in terms of x_0 and the given data $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. Assume σ^2 is known.
- Sketch the standard deviation of $\hat{\mu}_0$ as a function of x_0 . Indicate the position of $\bar{x}$ on the x-axis.

$$\begin{aligned} \text{Var}(\hat{\mu}_0) &= \text{Var}(\hat{\beta}_0) + x_0^2 \text{Var}(\hat{\beta}_1) + 2x_0 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ \text{Var}(\hat{\mu}_0) &= \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} + x_0^2 \left(\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) - 2x_0 \left(\frac{\sigma^2 \sum_{i=1}^n x_i}{n \sum_{i=1}^n (x_i - \bar{x})^2} \right) \end{aligned}$$

→ simplify:

→ define term G:

$$\begin{aligned} G &= \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2\bar{x} \sum_{i=1}^n x_i = \\ &= \sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2\bar{x}(n\bar{x}) = \sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2n\bar{x}^2 = \\ G &= \sum_{i=1}^n x_i^2 - n\bar{x}^2 \\ \text{also } (G + n\bar{x}^2) &= \sum_{i=1}^n x_i^2 \end{aligned}$$

$$\begin{aligned} \text{Var}(\hat{\mu}_0) &= \frac{\sigma^2 \sum_{i=1}^n x_i^2}{nG} + x_0^2 \left(\frac{\sigma^2}{G} \right) - 2x_0 \bar{x} \left(\frac{\sigma^2}{G} \right) \\ \text{Var}(\hat{\mu}_0) &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{G} \right) + x_0^2 \left(\frac{\sigma^2}{G} \right) - 2x_0 \bar{x} \left(\frac{\sigma^2}{G} \right) \\ \text{Var}(\hat{\mu}_0) &= \frac{\sigma^2}{n} + \frac{\sigma^2}{G} (\bar{x}^2 + x_0^2 - 2x_0 \bar{x}) \end{aligned}$$

$$\text{Var}(\hat{\mu}_0) = \frac{\sigma^2}{n} + \frac{\sigma^2}{G} (x_0 - \bar{x})^2$$